

Ethanol Blending in Gasoline – Germany

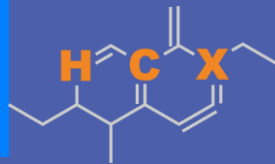
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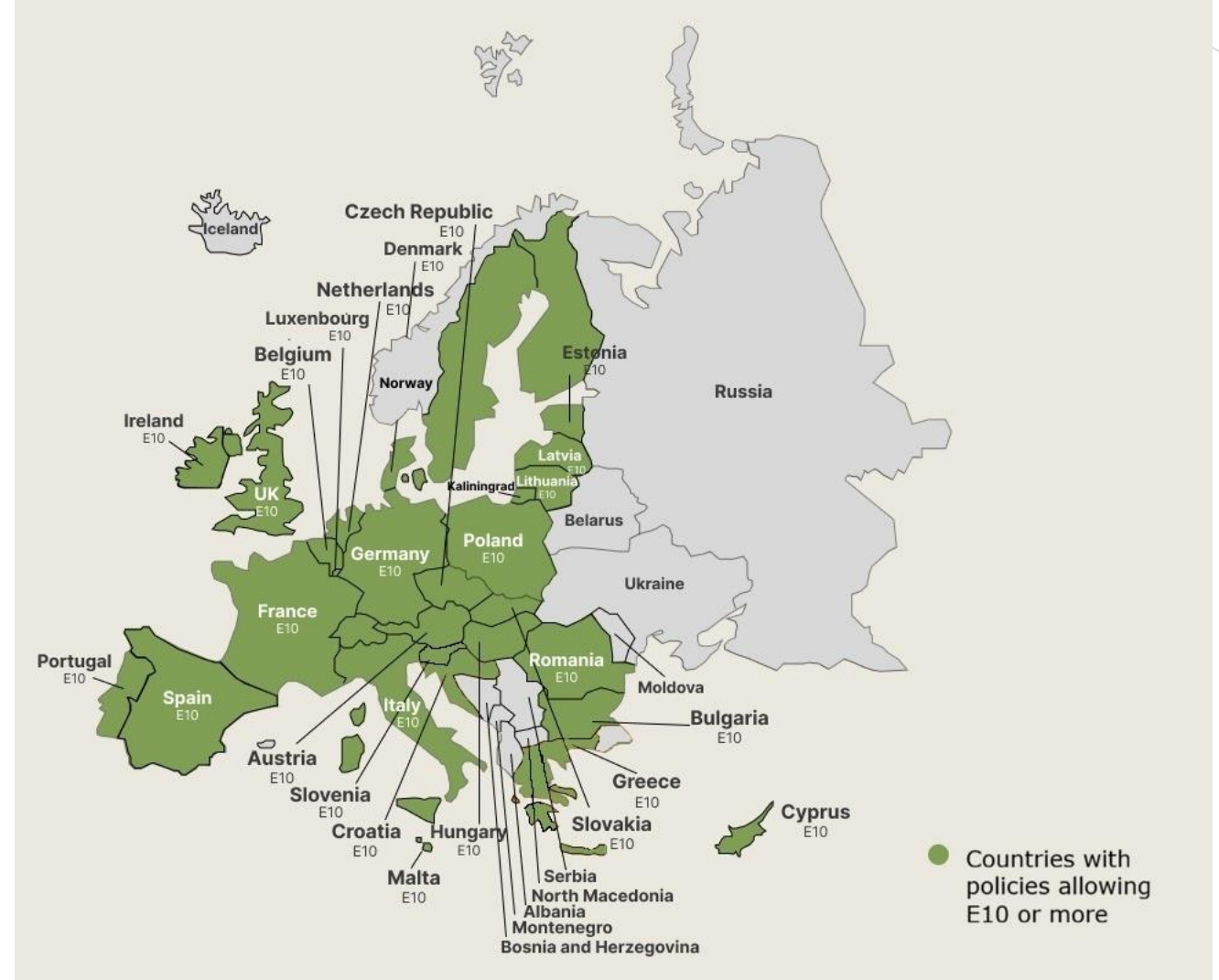


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Germany



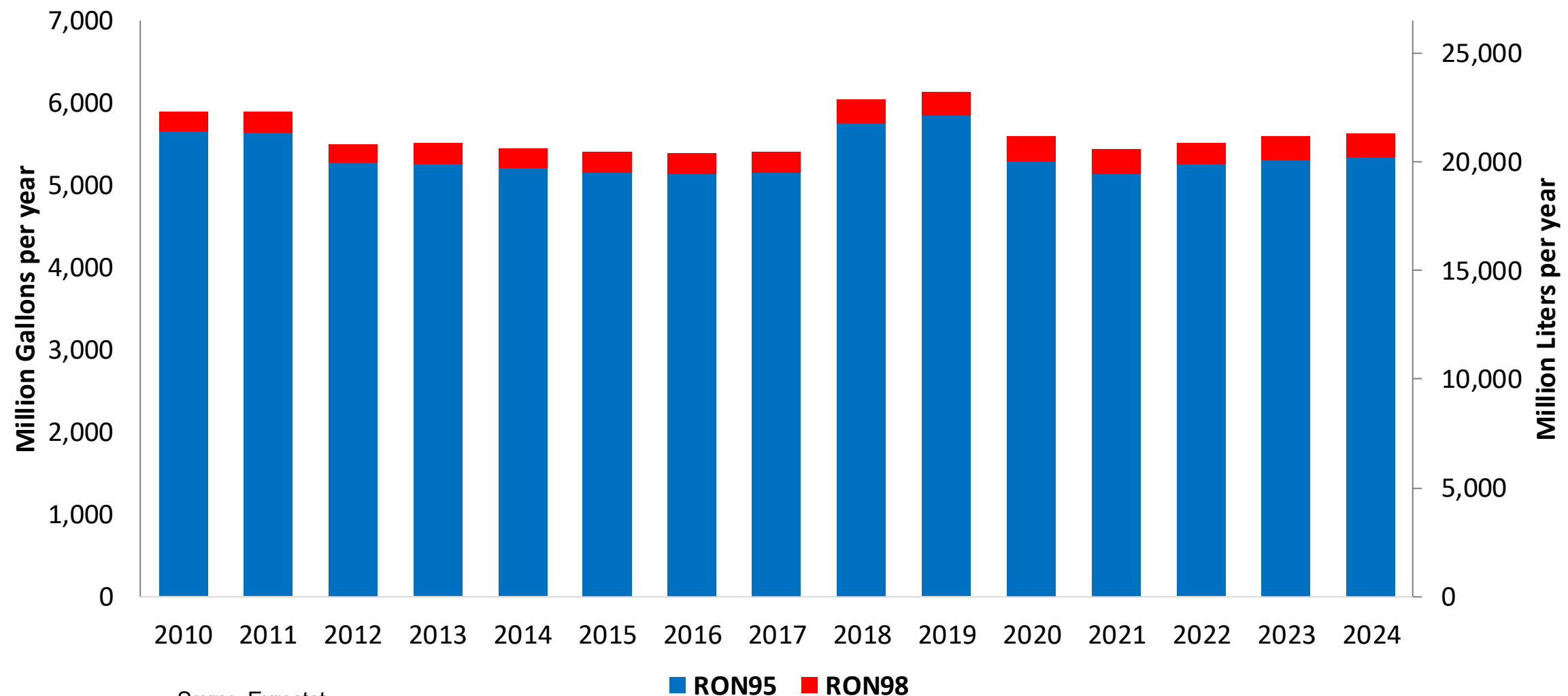
In 2024, gasoline consumption in Germany was 21,292 million liters (5,625 million gallons) and growth rate was -1.7%. Gasoline production and net imports were 22,284 and -3,611 million liters (5,887 and -954 million gallons) correspondingly. Germany complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2024, ethanol consumption in Germany was 2,262 million liters (598 million gallons) representing 10.6% of gasoline demand. Ethanol production and Net Imports were 1,333 and 929 million liters (598 and 245 million gallons) correspondingly. Ethanol sources are Cereals 64% and Sugar Beet 33%.

Source: Eurostat, European Commission

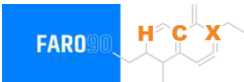
Gasoline Demand in Germany

Gasoline Demand by Grade in Germany

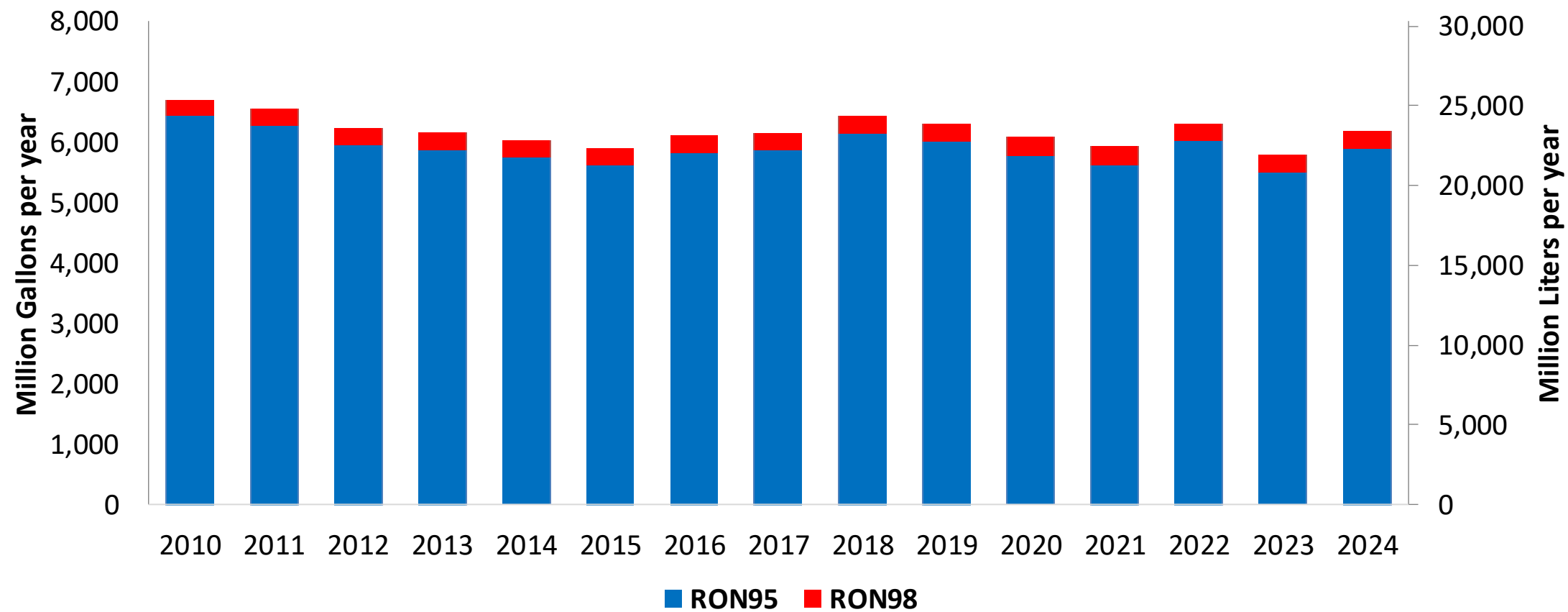


Source: Eurostat

Gasoline Production in Germany



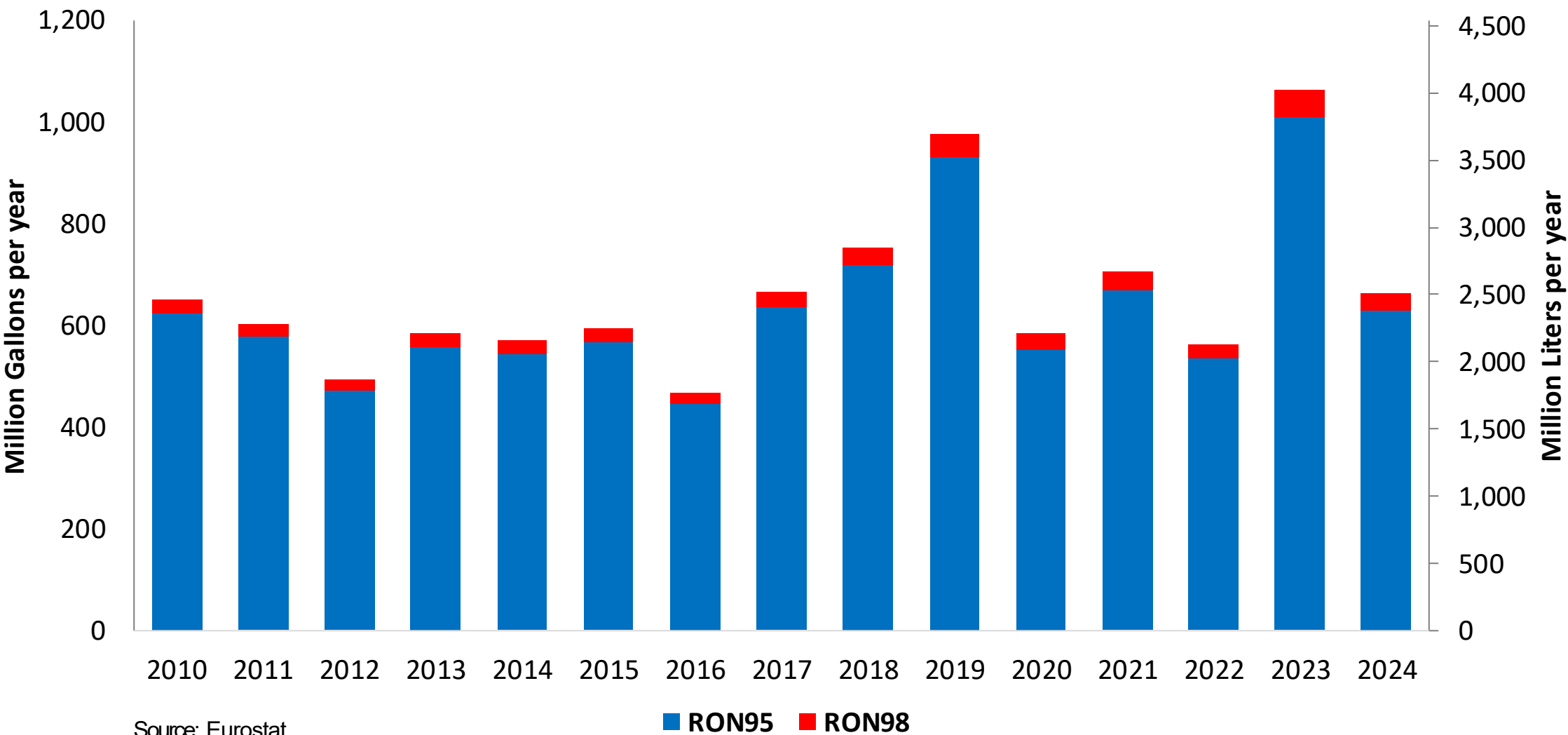
Gasoline Production by Grade in Germany



Source: Eurostat

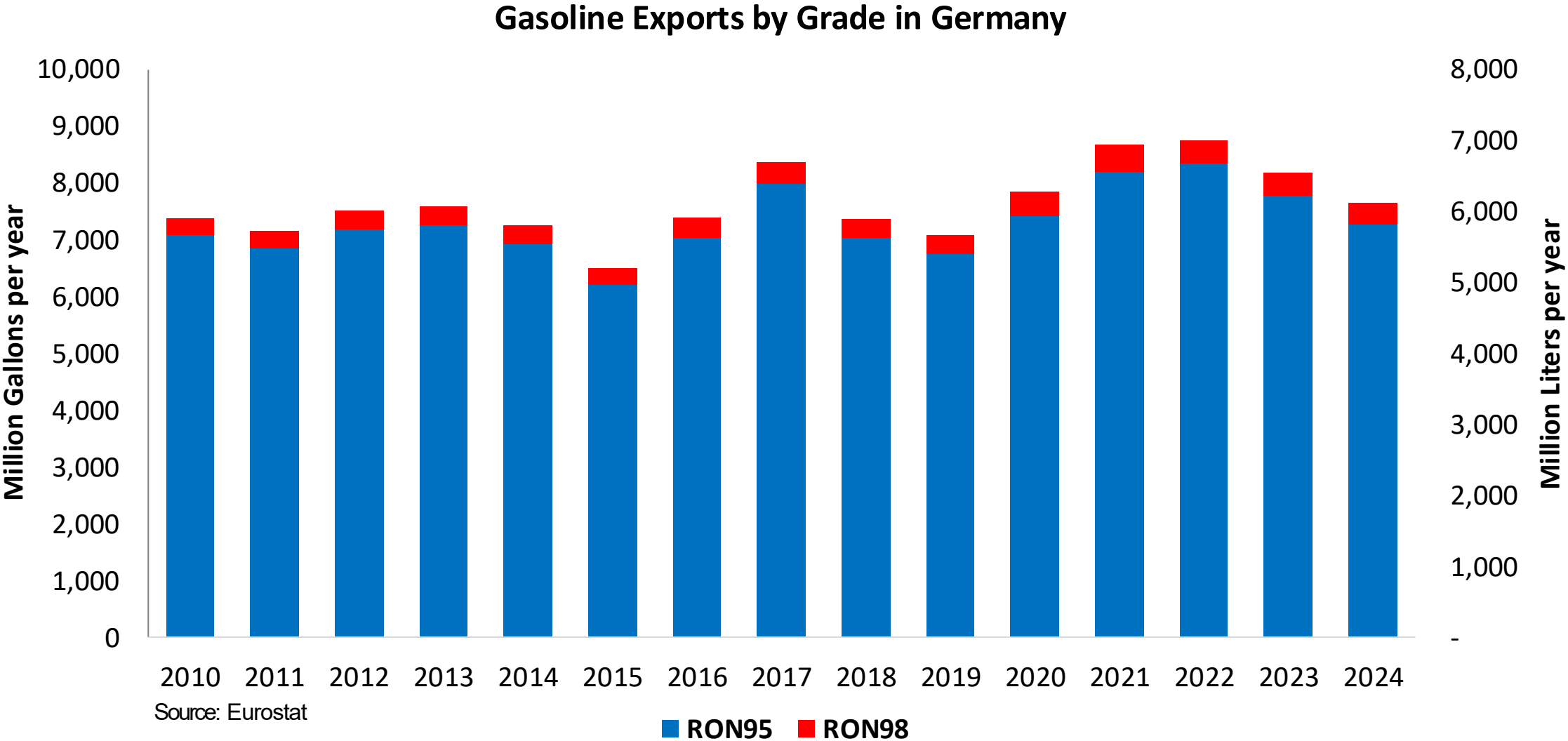
Gasoline Imports in Germany

Gasoline Imports by Grade in Germany

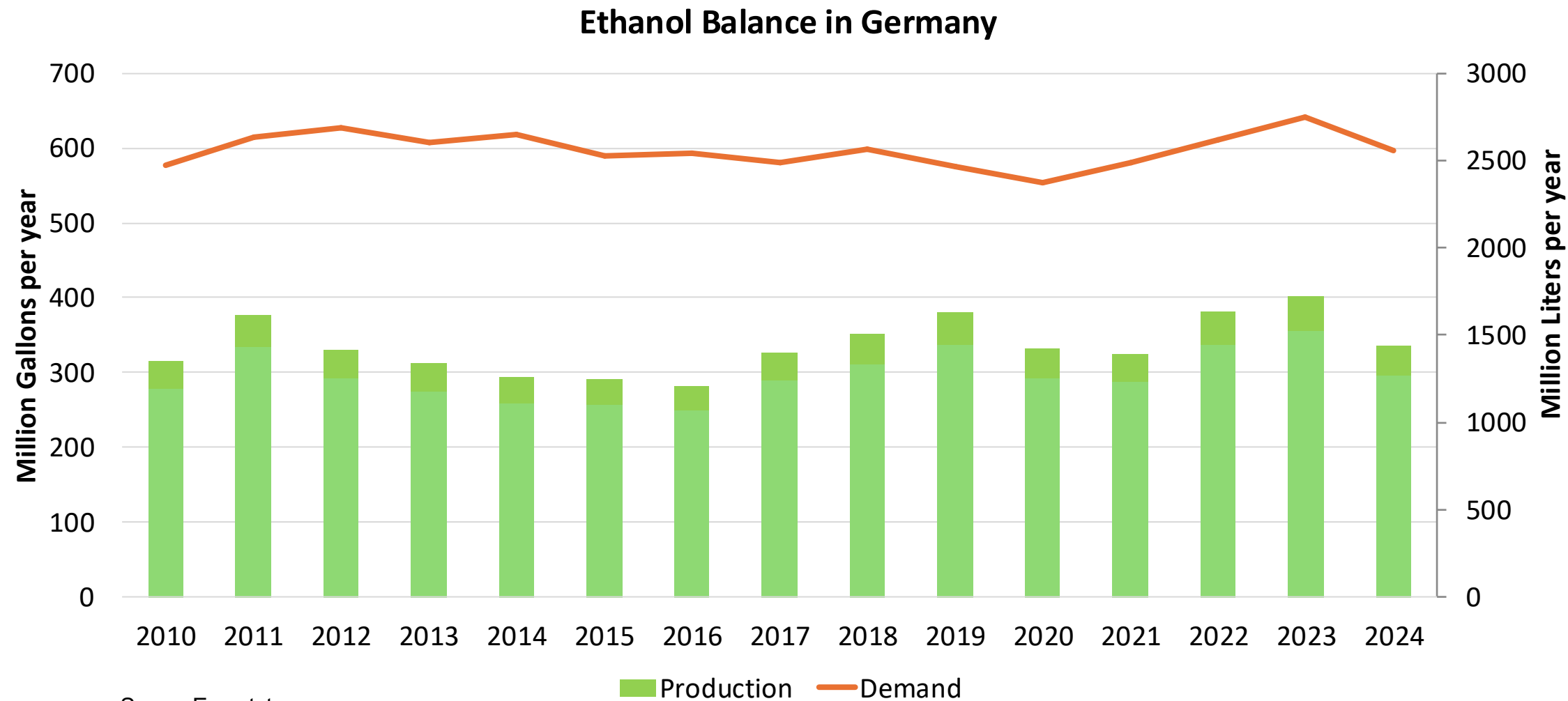


Source: Eurostat

Gasoline Exports in Germany



Ethanol Balance in Germany



Source: Eurostat

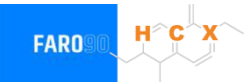
Gasoline Quality in Germany

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.7							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	10.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

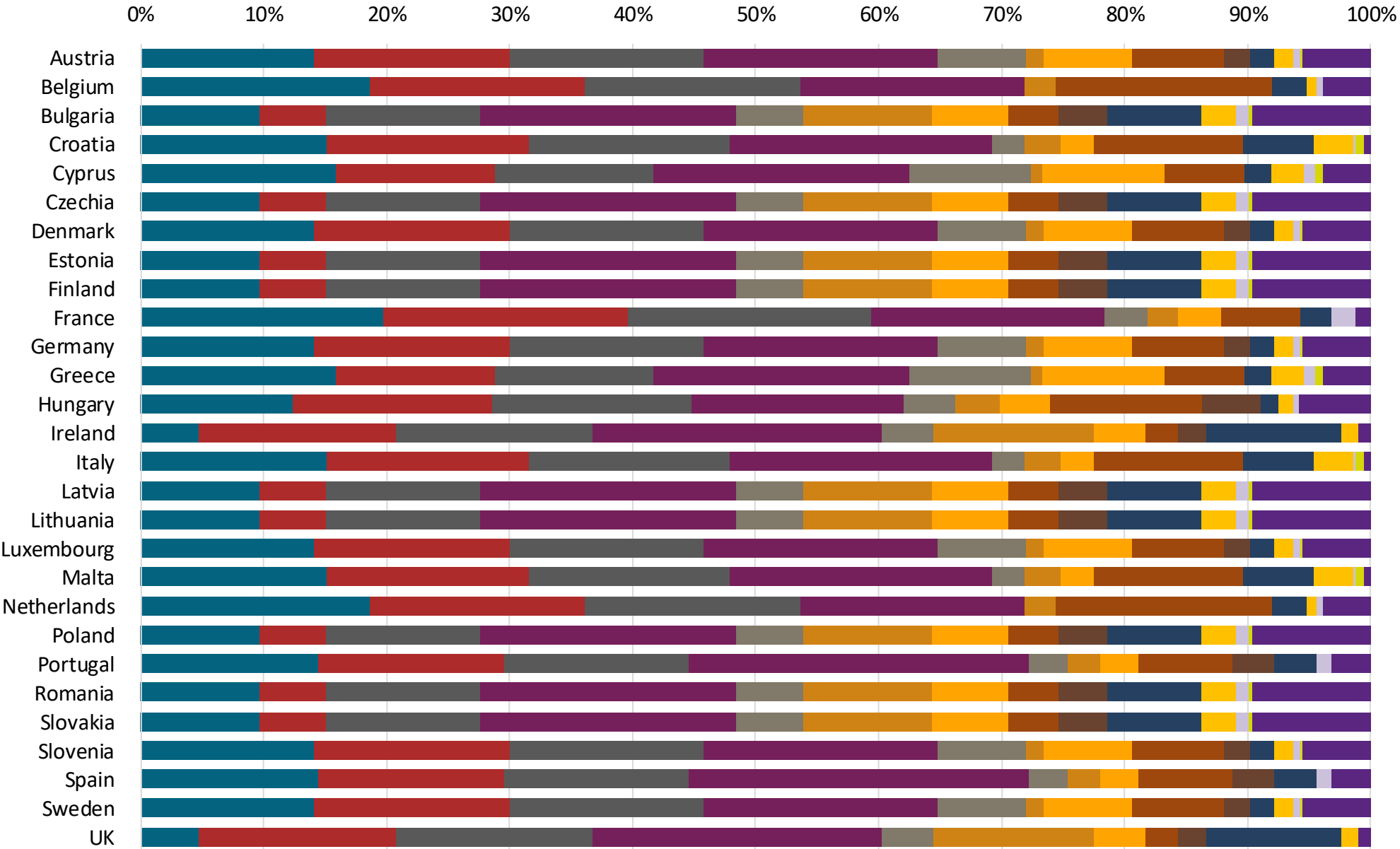
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



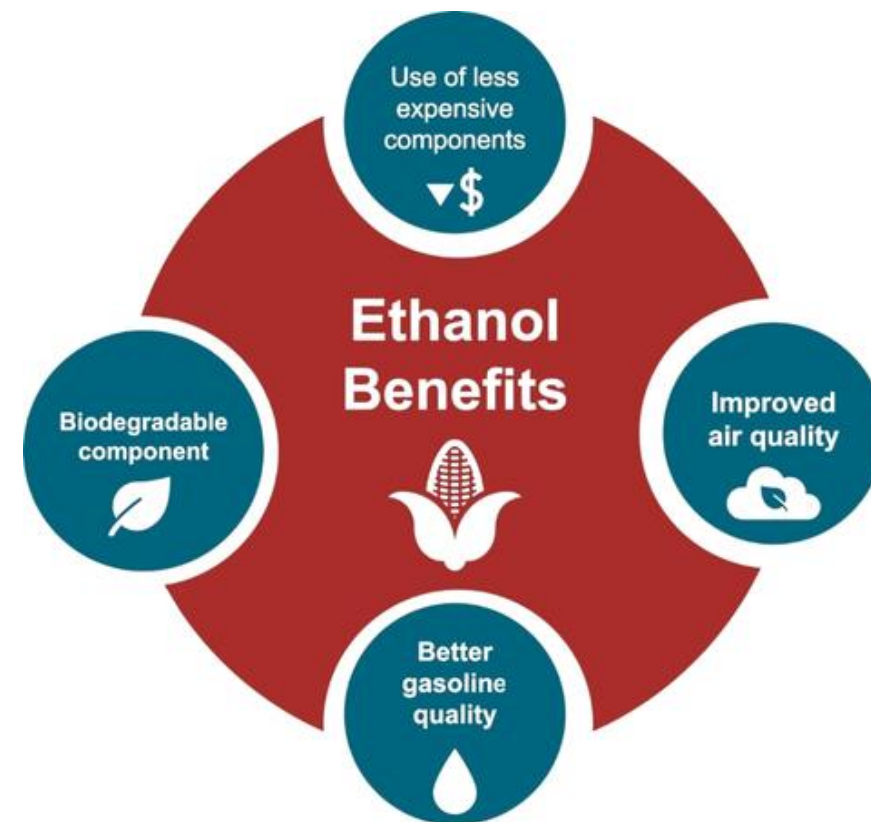
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

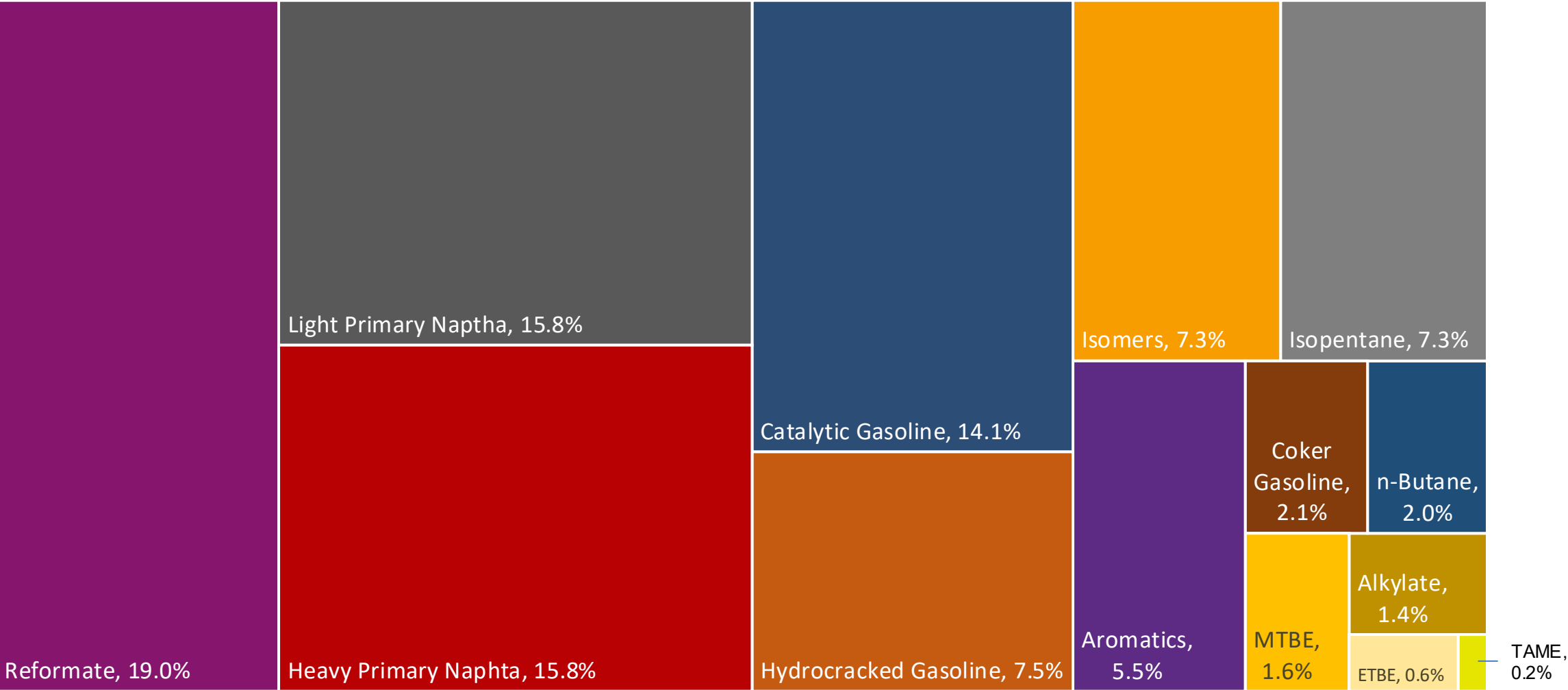
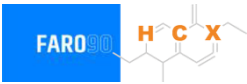
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

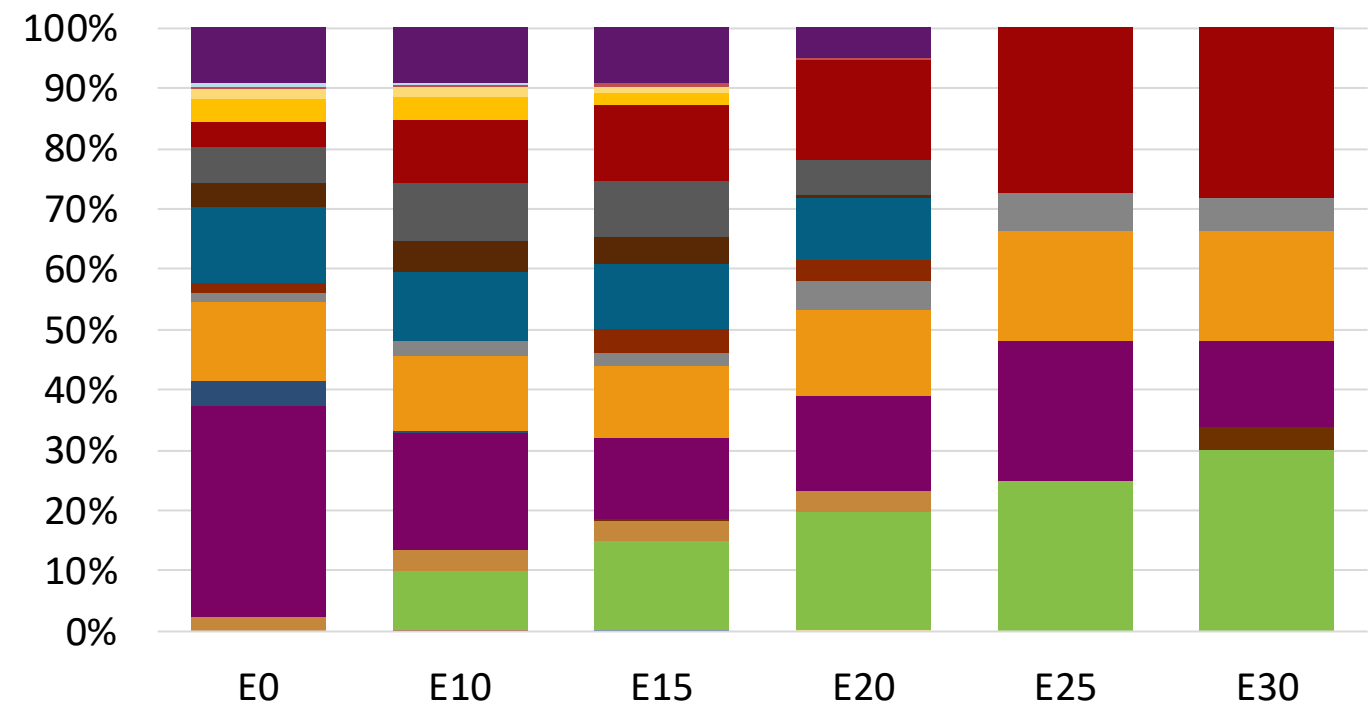
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Germany Available Blending Components



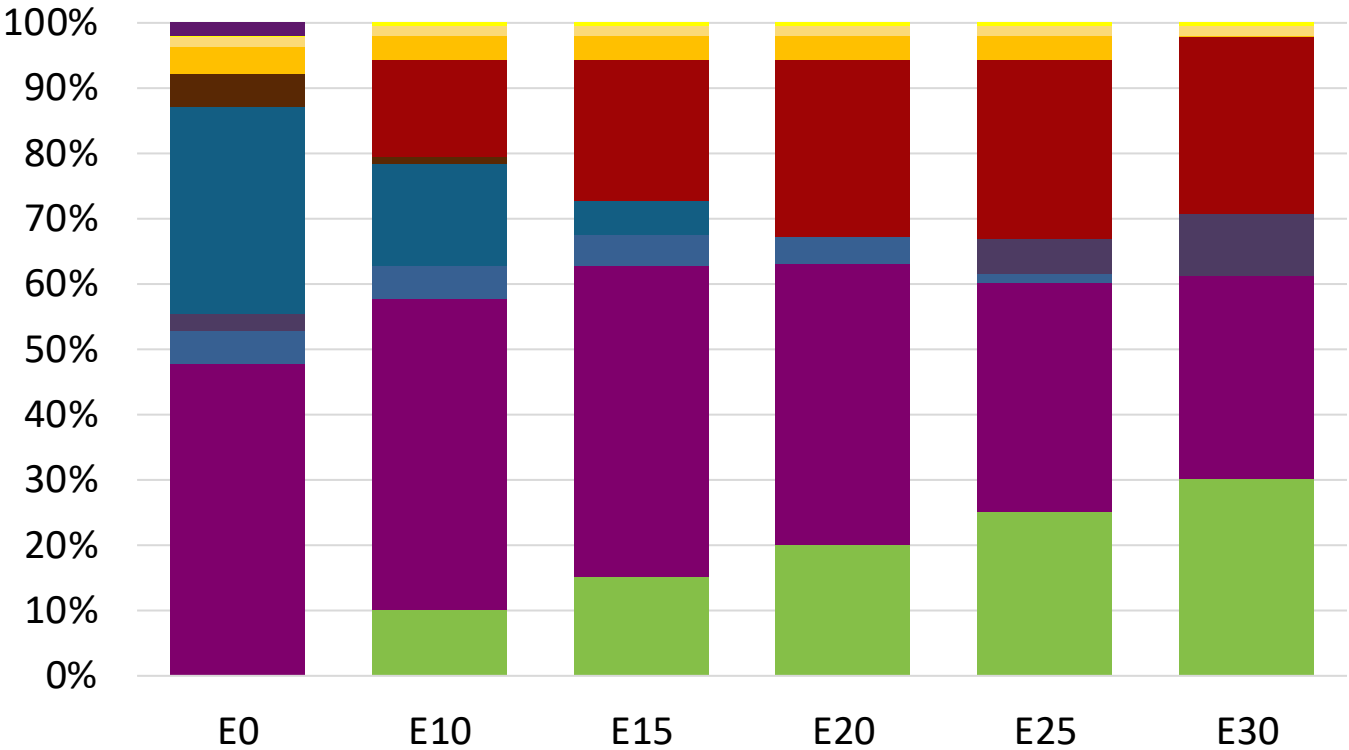
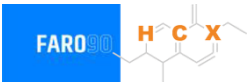
Germany – Gasoline 95 – Constant Octane



Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Germany - Gasoline 98 - Constant Octane

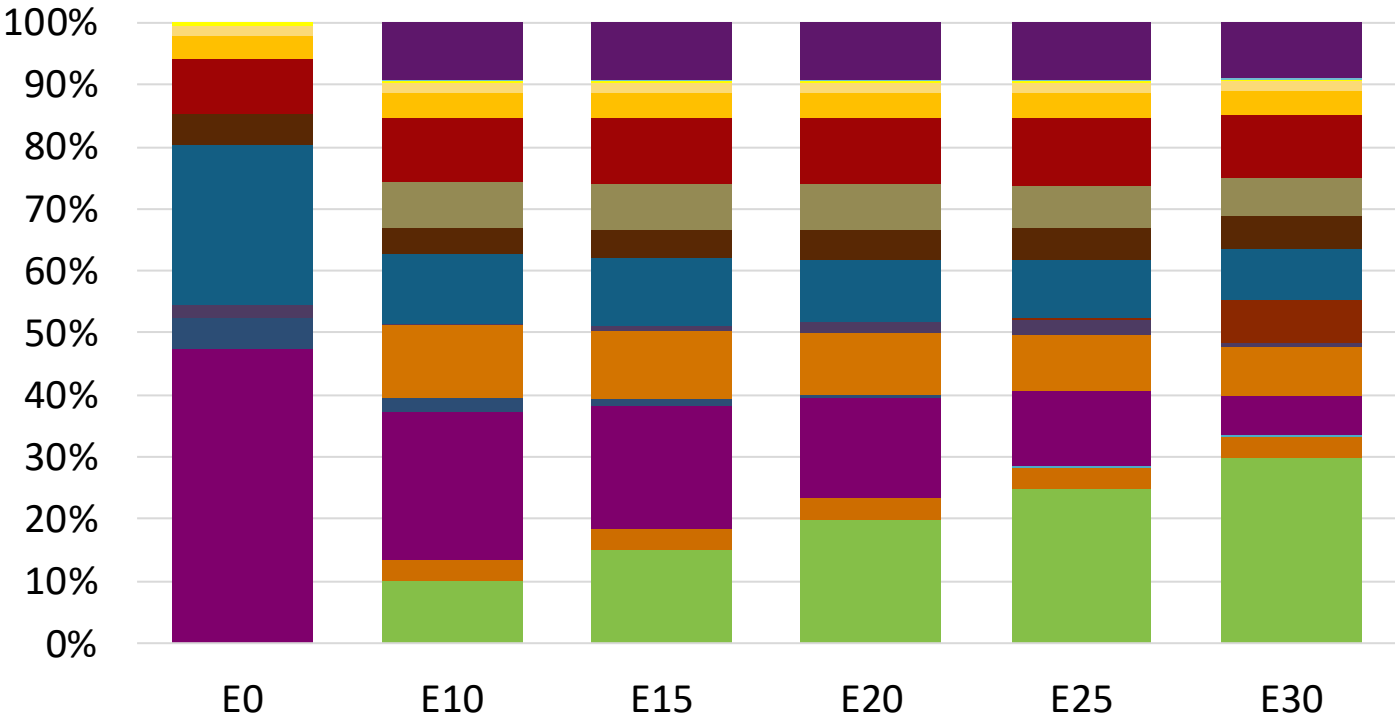


Average CIF 2024 Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

Source: Faro90

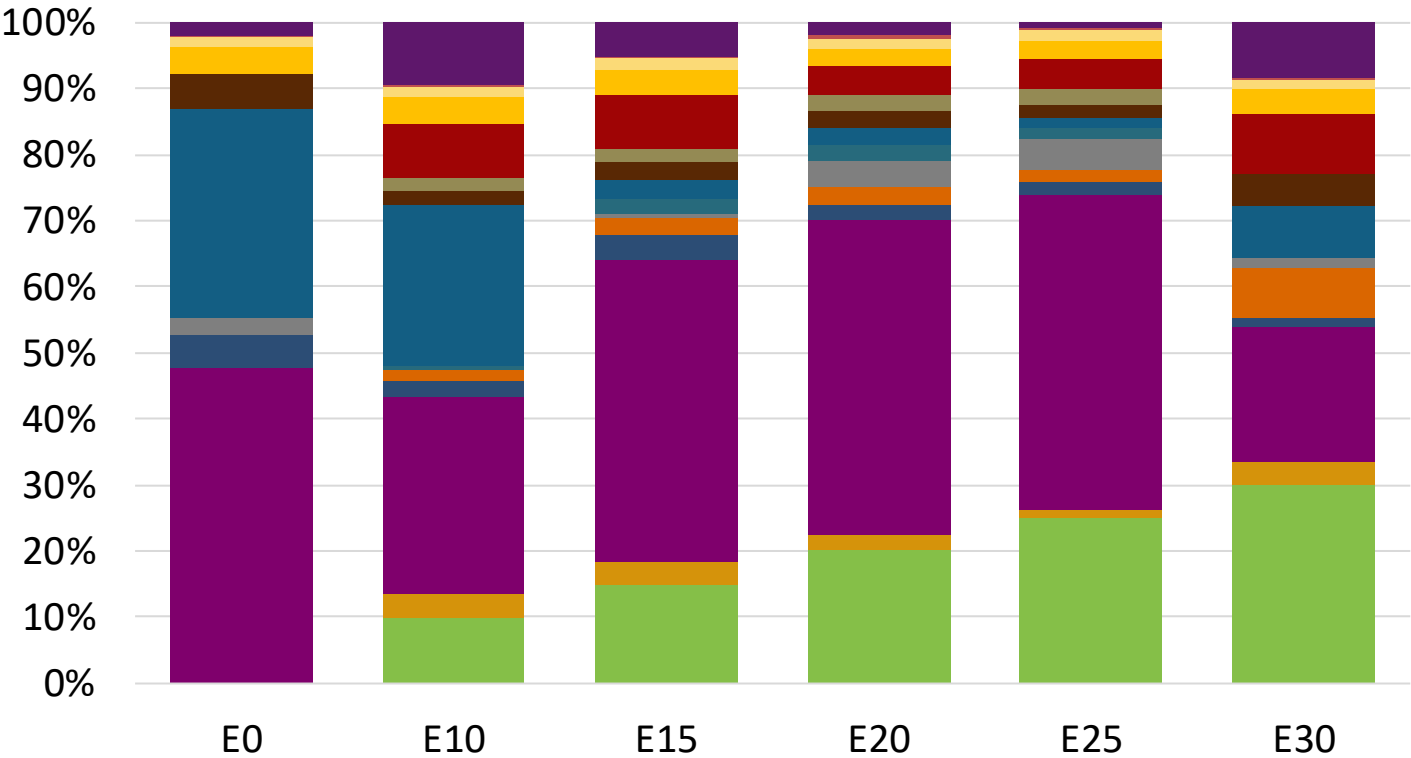
Germany – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Germany – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

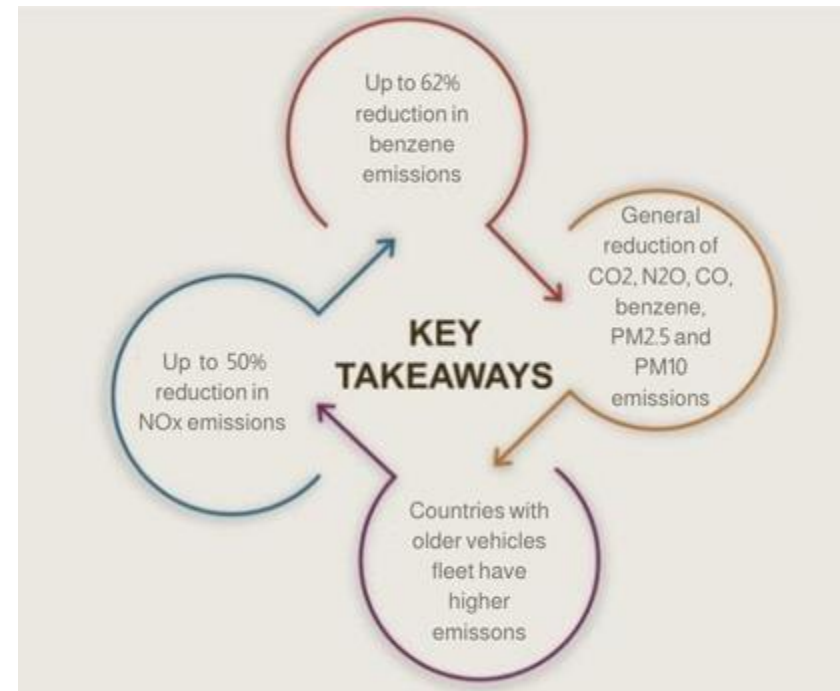
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

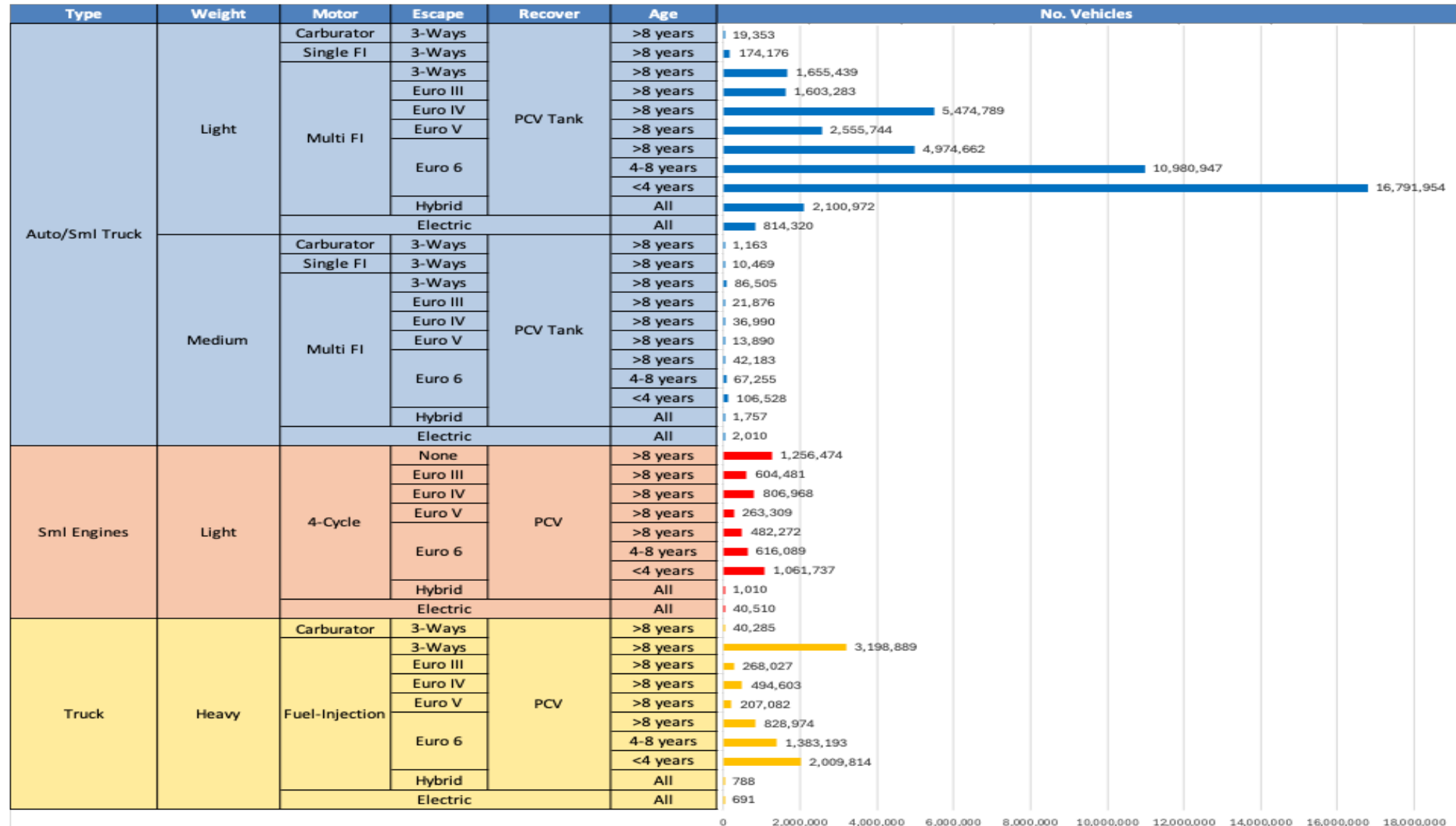
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Germany

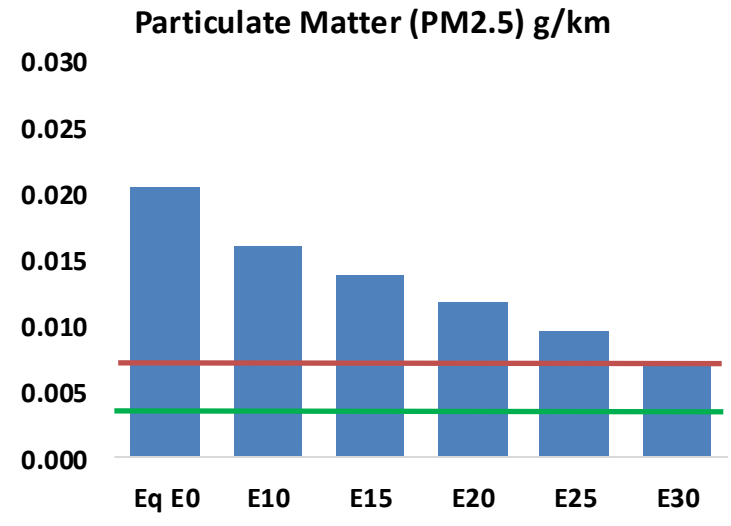
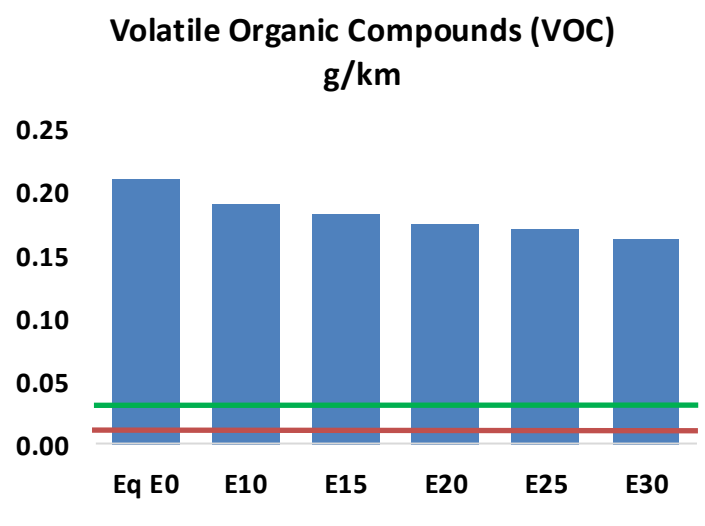
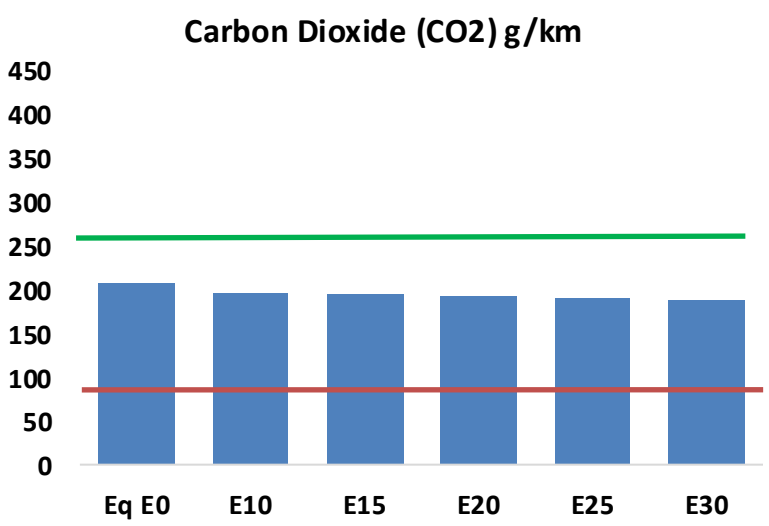
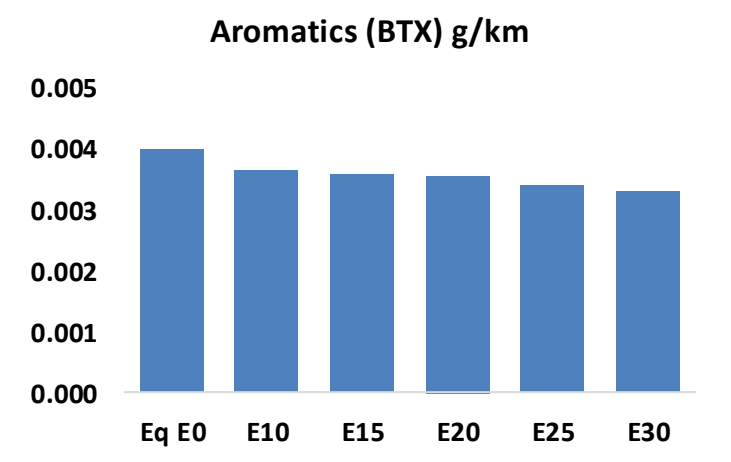
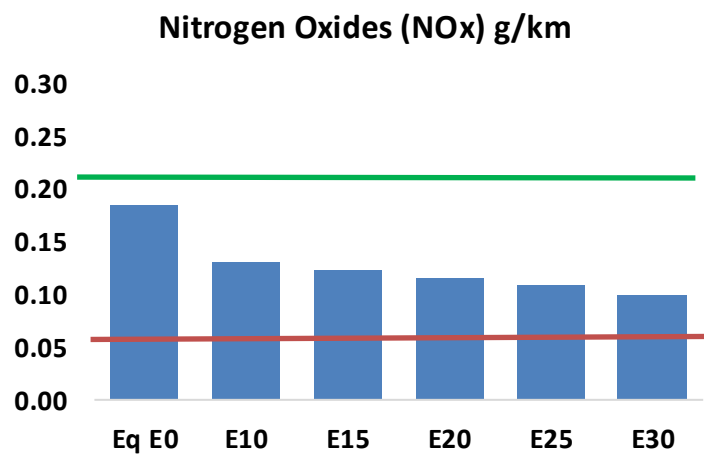
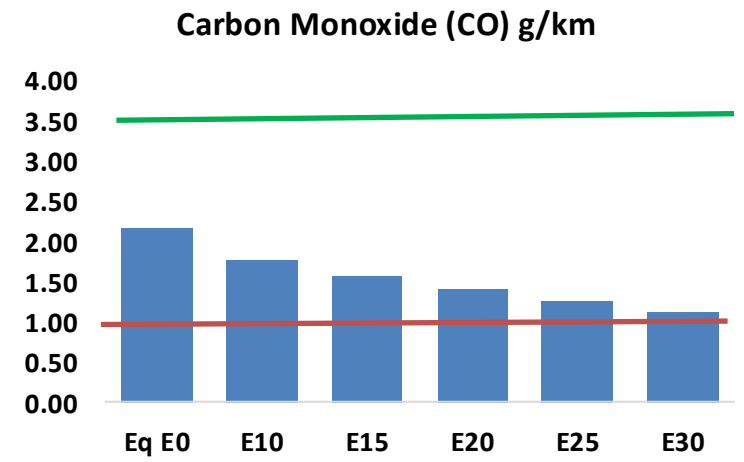
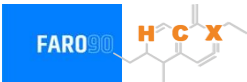


Vehicle Fleet: 61,101,463
Gasoline Vehicle Fleet: 32,331,793

Source: Eurostat, ACEA

Germany – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Germany – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,772,395	1,605,976	1,439,558	1,297,299	1,151,430	1,038,448	921,561	-19%	-35%
CO2	169,687,313	165,323,925	161,202,948	157,961,920	156,358,133	155,431,613	152,566,562	-5%	-8%
VOC	172,340	163,809	155,277	149,207	143,521	139,277	133,163	-10%	-17%
NOx	152,188	114,141	106,531	100,406	94,691	88,391	81,725	-30%	-38%
SOx	1,977	1,827	1,678	1,551	1,430	1,299	1,161	-15%	-28%
NH3	58,379	57,801	57,223	57,495	58,141	58,596	58,974	-2%	0%
N2O	1,820	1,810	1,803	1,811	1,849	1,869	1,891	-1%	2%
CH4	41,275	41,275	41,275	41,894	42,802	43,586	44,329	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,662	1,583	1,505	1,451	1,401	1,363	1,309	-9%	-16%
Acetaldehyde	3,372	3,870	5,679	8,648	11,783	13,464	15,909	68%	249%
Formaldehyde	10,885	10,583	12,337	14,486	15,176	16,789	18,277	13%	39%
Benzene	3,279	3,151	2,986	2,939	2,914	2,787	2,697	-9%	-11%
PM2.5	10,785	9,576	8,368	7,267	6,171	5,007	3,795	-22%	-43%
PM10	6,036	5,360	4,683	4,067	3,453	2,802	2,124	-22%	-43%

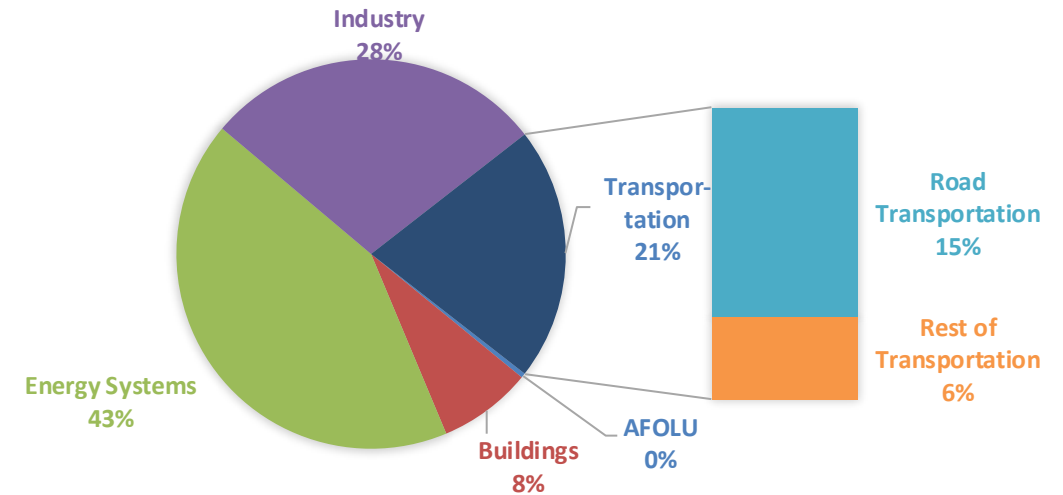
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

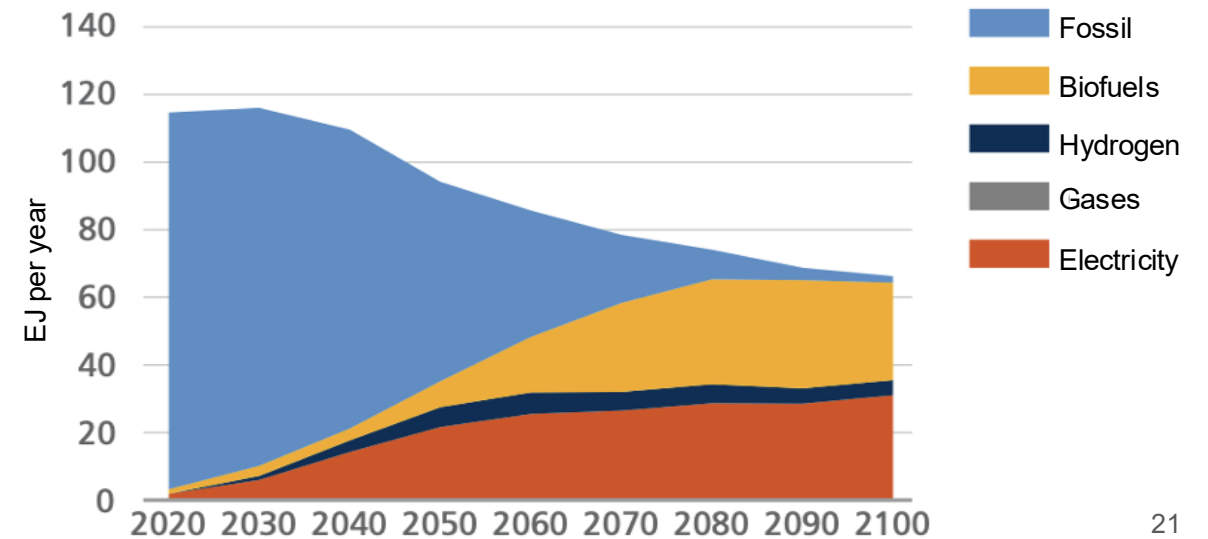
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

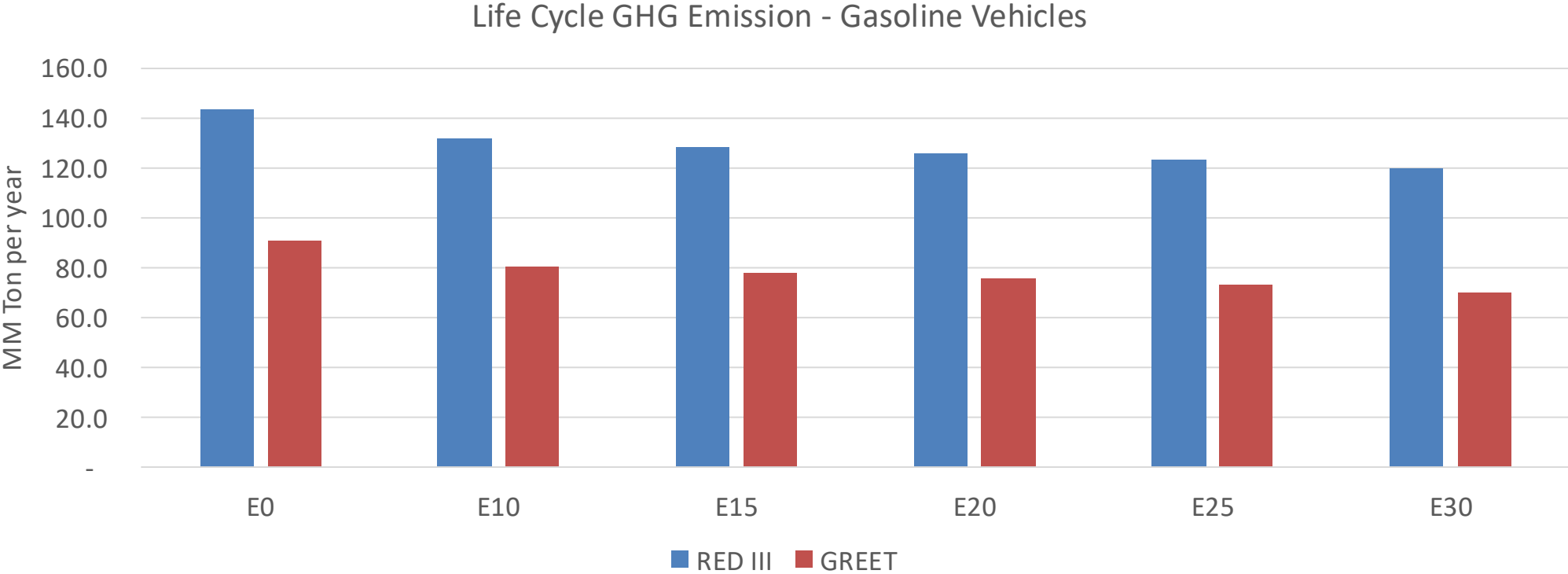
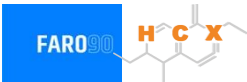
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Germany – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	717.1
Target @ 2035 (MMT/year)	403.6
Delta	313.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.4%	16.9%	19.3%	22.6%
RED II/III	0.0%	7.9%	10.3%	12.2%	14.1%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.2%	4.2%	4.9%	5.6%	6.5%
RED II/III	0.0%	3.6%	4.7%	5.6%	6.4%	7.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90